

Mathematics for Data Science

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Motivation

- **Uncertainty Modeling:** Probability allows machine learning models to quantify and manage uncertainty in predictions and data, enabling better decision-making and risk assessment.
- **Bayesian Inference:** Conditional probability is fundamental in Bayesian inference, which provides a framework for updating model beliefs based on new evidence, improving model accuracy and adaptability.
- **Classification and Prediction:** Probabilistic models, such as Naive Bayes and probabilistic graphical models, rely on probability and conditional probability to make predictions and classifications, allowing for robust and interpretable outcomes.

Random Experiment

- Several processes are random in nature, where the total possible outcomes are known but the occurrence of an outcome is uncertain. Such processes or experiments are called *random experiments*.
- Example: tossing a coin, rolling a dice, etc.
- Probability is the *measure of uncertainty* of such a random experiment.

Classical Approach to Probability

$$\text{Probability} = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}}$$

Definitions

1. **Sample Space:** The set S of all possible outcomes of a random experiment.
2. **Event:** Any subset E of the sample space ($E \subseteq S$).
3. **Types of Events:**
 - **Sure Event:** Event which is certain to occur, $P(E) = 1$.
 - **Impossible Event:** Event which is impossible to occur, $P(E) = 0$.
 - **Mutually Exclusive Events:** Events E and F are mutually exclusive if they cannot occur simultaneously, $E \cap F = \phi$.
 - **Exhaustive Events:** Events E and F are exhaustive events if they exhaust the whole sample space, that is, $E \cup F = S$.

Example

$$S = \{1, 2, \dots, 100\}$$

E_1 = event of getting odd numbers

$$= \{1, 3, \dots, 99\}$$

E_2 = event of getting even numbers

$$= \{2, 4, \dots, 100\}$$

$E_1 \cap E_2 = \phi \implies E_1$ and E_2 are mutually exclusive events.

$E_1 \cup E_2 = S \implies E_1$ and E_2 are mutually exhaustive events.

Addition Rule on Probability

For any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For any three events A , B and C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ + P(A \cap B \cap C)$$

Example: One integer is chosen at random from the integers $1, 2, \dots, 100$. What is the probability that the chosen integer is divisible by 6 or 8?

Solution

$$S = \{1, 2, \dots, 100\}$$

E_1 = event of getting an integer divisible by 6

$$= \{6, 12, \dots, 96\}$$

E_2 = event of getting an integer divisible by 8

$$= \{8, 16, \dots, 96\}$$

$$n(S) = 100, n(E_1) = 16, n(E_2) = 12$$

$$E_1 \cap E_2 = \{24, 48, 72, 96\}, n(E_1 \cap E_2) = 4$$

$$P(E_1 \text{ or } E_2) = P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$= \frac{16}{100} + \frac{12}{100} - \frac{4}{100} = \frac{24}{100} = 0.24$$

Conditional Probability

For any two events E and F associated with a sample space S of a random experiment, the probability of event E given event F has occurred is called as conditional probability of E given F , denoted as $P(E|F)$.

$$P(E|F) = \frac{P(E \cap F)}{P(F)}; \text{ provided } P(F) \neq 0$$

Example: A black and a red die are rolled. What is the conditional probability of obtaining a sum greater than 9 given that the black die result in a 5?

Solution

E_1 = event of getting a 5 in a black coloured die

E_2 = event of getting sum greater than 9

$$E_1 = \{(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)\}$$

$$E_1 \cap E_2 = \{(5, 5), (5, 6)\}$$

$$P(E_2|E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{\frac{2}{36}}{\frac{6}{36}} = \frac{2}{6} = \frac{1}{3}$$

Conditional Probability: Properties

1. For any non-empty event E , $P(E|E) = P(S|E) = 1$
2. For any events A , B , and $F \neq \phi$,

$$P((A \cup B)|F) = P(A|F) + P(B|F) - P((A \cap B)|F)$$

3. For any events A , and $F \neq \phi$, $P(A^c|F) = 1 - P(A|F)$

Problem: A problem is given to two students: A and B. The probability that student A can solve the problem given that student B can solve it is $\frac{3}{7}$ and the probability that student B can solve the problem given that student A cannot solve it is $\frac{1}{7}$. If the probability that student B can solve the problem is $\frac{1}{10}$, then what is the probability that student A can solve it?

Solution

$$P(B) = \frac{1}{10}$$

$$P(A|B) = \frac{3}{7} = \frac{P(A \cap B)}{P(B)} \implies P(A \cap B) = \frac{3}{70}$$

$$P(B|A') = \frac{1}{7} = \frac{P(A' \cap B)}{P(A')} = \frac{P(B) - P(A \cap B)}{1 - P(A)}$$

$$\implies \frac{\frac{1}{10} - \frac{3}{70}}{1 - P(A)} = \frac{1}{7} \implies 1 - P(A) = \frac{2}{5}$$

$$\implies P(A) = \frac{3}{5}$$

Multiplication Theorem on Probability

For Two Events: For non-empty events E and F associated with sample space S of a random experiment, the multiplication rule states that the probability of simultaneous occurrence of the events E and F is

$$P(E \cap F) = P(E) \cdot P(F|E) = P(F) \cdot P(E|F); \text{ provided } P(E), P(F) \neq 0$$

For Three Events:

$$P(E \cap F \cap G) = P(E) \cdot P(F|E) \cdot P(G|E \cap F); \text{ provided } P(E), P(F), P(G) \neq 0$$

Example: Three cards are drawn simultaneously without replacement from a deck of 52 cards. What is the probability that the first two cards are king and the third card drawn is a queen?

Solution

E_1 = event of getting first card as king

E_2 = event of getting second card as king

E_3 = event of getting third card as queen

$$\begin{aligned} P(E_1 \cap E_2 \cap E_3) &= P(E_1) \cdot P(E_2|E_1) \cdot P(E_3|E_1 \cap E_2) \\ &= \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{4}{50} \end{aligned}$$

Independent Events

Two events E and F are said to be independent if the occurrence of one does not effect the occurrence of the other. We say events E and F are independent if:

$$P(E \cap F) = P(E) \cdot P(F) \text{ or}$$

$$P(E|F) = P(E); P(F) \neq 0 \text{ and } P(F|E) = P(F); P(E) \neq 0$$

Partition of a Sample Space

$\{E_1, E_2, \dots, E_n\}$ is said to be a partition of the sample space S if

- E_i 's are exhaustive events: $E_1 \cup E_2 \dots \cup E_n = S$
- E_i 's are mutually exclusive events: $E_i \cap E_j = \phi$, for $i \neq j$

Theorem of Total Probability

Consider $\{E_1, E_2, \dots, E_n\}$ to be a partition of the sample space S associated with a random experiment. For any event $A \subset S$, the total probability of A is given as

$$P(A) = P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + \dots + P(E_n) \cdot P(A|E_n)$$

$$= \sum_{j=1}^n P(E_j) \cdot P(A|E_j)$$

Example

A person has taken a construction job. The probabilities are 0.6 that there will be a strike, 0.8 that the construction job will be completed on time if there is no strike and 0.3 that the construction work will be completed on time if there is a strike. Determine the probability that the construction job will be completed on time.

Solution

A is the event that the job will be completed on time.

$P(E_1)$ is the probability that there is a strike

$P(E_1) = 0.6$, $P(E'_1) = 0.4$

$$\begin{aligned} P(A) &= P(E_1) \cdot P(A|E_1) + P(E'_1) \cdot P(A|E'_1) \\ &= 0.6 \cdot 0.3 + 0.4 \cdot 0.8 = 0.18 + 0.32 = 0.50 \end{aligned}$$

Thank You

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